

Quality control of hourly rain gauge data based on radar and satellite multi-source data

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ABSTRACT

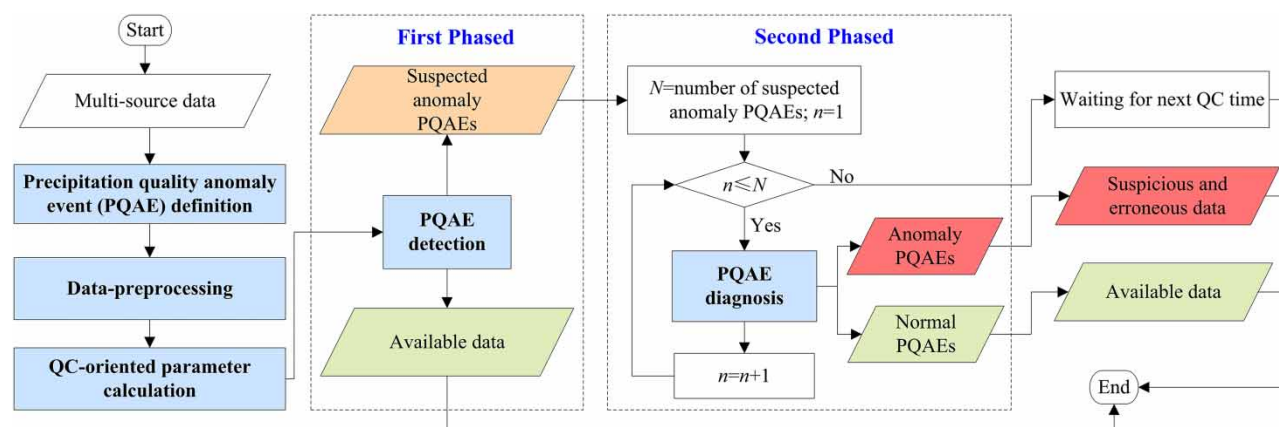
Rain gauge networks provide direct precipitation measurements and have been widely used in hydrology, synoptic-scale meteorology, and climatology. However, rain gauge observations are subject to a variety of error sources, and quality control (QC) is required to ensure the reasonable use. In order to enhance the automatic detection ability of anomalies in data, the novel multi-source data quality control (NMQC) method is proposed for hourly rain gauge data. It employs a phased strategy to reduce the misjudgment risk caused by the uncertainty from radar and satellite remote-sensing measurements. NMQC is applied for the QC of hourly gauge data from more than 24,000 hydro-meteorological stations in the Yangtze River basin in 2020. The results show that its detection ratio of anomalous data is 1.73%, only 1.73% of which are suspicious data needing to be confirmed by experts. Moreover, the distribution characteristics of anomaly data are consistent with the climatic characteristics of the study region as well as measurement and maintenance modes of rain gauges. Overall, NMQC has a strong ability to label anomaly data automatically, while identifying a lower proportion of suspicious data. It can greatly reduce manual intervention and shorten the impact time of anomaly data in the operational work.

Key words: multi-source data, precipitation, quality control, radar, satellite

HIGHLIGHTS

- Quantitative and fusion application of radar and satellite data were addressed for the quality control (QC) of hourly rain gauge data.
- Precipitation quality anomaly events were defined to improve the traceability of QC.
- A phased QC strategy was adopted to reduce the misjudgment risk caused by the uncertainty of remote-sensing measurements.

GRAPHICAL ABSTRACT



1. INTRODUCTION

Precipitation data are essential for numerous operational applications in hydrology, synoptic-scale meteorology, and climatology such as hydrological modeling (Mohammadi *et al.* 2024), weather forecasts (Imhoff *et al.* 2023), flood warning (Ma *et al.* 2024), drought forecasting (Mohammadi 2023), and decision-making service (Hassani *et al.* 2023). Precipitation data can be obtained by different means, e.g., rain gauge station, radar, and satellite. Among them, rain gauges provide direct precipitation measurements, which are routinely used as ground truth; radar and satellite precipitation, by contrast, are obtained based on the modern remote-sensing techniques, which have to be calibrated against the rain gauge data (Nešpor & Sevruk 1999; Martinaitis 2008; Qi *et al.* 2016; WMO 2021). In general, the direct precipitation measurements provided by gauge networks have higher accuracy than remote-sensing measurement systems. However, rain gauge observations are also subject to inaccuracies caused by random and systemic errors, and the main causes include wind-induced error, wetting and evaporation losses, instrumentation malfunctions, transmission errors, poor observation environment, and mistakes made during data processing (Groisman & Legates 1994; Nešpor & Sevruk 1999; Nešpor *et al.* 2000; Adam & Lettenmaier 2003; Yang *et al.* 2005; Baltas *et al.* 2016).

Quality control (QC) is the best known component of quality management systems to ensure the highest possible reasonable standard of accuracy for the optimum use of these data by all possible users (Zahumenský 2004). Although many sophisticated QC procedures have been carried out in various hydrological and meteorological research projects (Ren *et al.* 2010; Qi 2015; Blenkinsop *et al.* 2017), QC of rain gauge data has been a challenge, especially at hourly or sub-hourly scales, because of their high spatial and temporal variability with skewed intensity spectra (Li & Sun 2021; Sha *et al.* 2021).

In this study, the existing QC procedures are divided into two types: single-source QCs and multi-source QCs. Single-source QCs mainly use data from the checked rain gauge station or their neighbor stations, including limit value checks, station or regional extreme value checks, internal consistency checks, time consistency checks, and spatial consistency checks (Upton & Rahimi 2003; Kondragunta & Shrestha 2006; Ren *et al.* 2010; Schneider *et al.* 2014; Blenkinsop *et al.* 2017). Although these procedures have played an important role in many projects, the labeled suspected anomalous data are mostly required to make a further confirmation by human visual checks. With the continued growth of data volume, these procedures are resource-intensive and can cause delays. Meanwhile, radar and satellite products are being used in an increasing number of applications because of their wide spatial and temporal coverage (Lengfeld *et al.* 2020; Adane *et al.* 2021; Liu *et al.* 2021; Thiruvengadam *et al.* 2021; Gebremicael *et al.* 2022; Zhao *et al.* 2022). On the basis of this, some multi-source QCs have been developed to increase the efficiency of gauge precipitation QC by introducing data from different observation systems (Hill 2013; Qi & Zhang 2013; Qi 2015; Qi *et al.* 2016; Zhao *et al.* 2018; Sha *et al.* 2021), and some new technologies have been adopted for automatic labeling of the anomalies in data, e.g. decision trees (Qi *et al.* 2016), neural networks (Zhao *et al.* 2018), deep learning (Sha *et al.* 2021), but analysis shows that the magic of these methods is inseparable from the support of multi-source data.

The existing multi-source QCs mainly focus on the use of radar data, and there are few studies on the comprehensive application of radar and satellite. The accuracy of ground-based radar data is easily affected by complex terrain and its usage is limited in areas of poor or no radar coverage. However, satellites, especially geostationary satellites, can prove to be an excellent data source providing high spatial and temporal resolutions for regions where radar networks are missing or unevenly distributed. In this paper, a novel multi-source data quality control method called NMQC is proposed for the hourly rain gauge data by the quantitative and fusion application of radar and satellite data. Notably, although NMQC is designed to be as automatic as possible to improve timeliness and reduce the human workload, it is not fully automatic owing to the skewness of precipitation frequency distribution. The contributions are summarized as follows:

- Precipitation quality anomaly event (PQAE) is defined with massive data exploration and analysis to improve the traceability management of QC, and several types of QC-oriented parameters are designed to support the implementation of related algorithms based on the multi-source data.
- A phased strategy is applied to logically divide QC procedures into two steps of PQAE detection and PQAE diagnosis to reduce the misjudgment risk caused by the uncertainties from radar and satellite remote-sensing measurements.
- A hybrid QC processing mode of automatic and human-based is reserved to avoid the incorrect elimination of rare extreme precipitation values occurring in the fully automatic QC procedures, which behave similarly to spurious outliers, as these true extremes are very important to describe the variability of precipitation.

2. METHODS

2.1. Study area description

The study region is the Yangtze River Basin (24.50°–35.75°N, 90.55°–122.42°E) with a total area of ~1.8 million km², accounting for nearly 19% of China's land area. With a length of over 6,300 km, the Yangtze River is the longest river in Asia and the third longest river in the world. The Yangtze River Basin stretches from the eastern Tibetan Plateau to the East China Sea with a wide range of climate variability and diverse ecosystems (Zhang *et al.* 2014). Affected by its unique geographic location, special land-sea thermal differences, and seasonal variations of atmospheric circulation, the Yangtze River Basin is a typical East Asian monsoon region that is sensitive and vulnerable to climate changes, characterized by simultaneous rainy and hot weather in summer, and cool and dry conditions in winter. In recent years, against the background of climate warming, the annual precipitation in the Yangtze River Basin shows a trend of dry and wet polarization, and extreme precipitation events occur frequently, which brings new challenges to the QC of precipitation data (Lin *et al.* 2021; Cheng *et al.* 2022).

Three types of data are adopted in this study. The station data are from more than 24,000 automated hydro-meteorological observation stations, as shown in Figure 1, including the hourly precipitation, air temperature, weather phenomena, and station metadata. Satellite data are a set of L2 processing level nominal grid products obtained from the FengYun-4A (FY-4A), which is the first flight unit of the second-generation geostationary meteorological satellite in China (Lu *et al.* 2017; Shao *et al.* 2020), including Cloud Top Temperature (CTT), CCloud Type (CLT), and CCloud Mask (CLM). FY-4A products are updated every 15 min (or intensive observation of 7 times per hour) at a spatial resolution of 0.04° × 0.04°. Moreover, Cloud Total Amount (CTA) product is used for auxiliary cloud parameters, which are produced by China's 3D Cloud Analysis System and updated every 1 h at a spatial resolution of 0.05° × 0.05° (Shi *et al.* 2019). Radar data comprise the Composite Reflectivity (CR) product, which is produced by Severe Weather Automatic Nowcasting System and updated every 6 min at a spatial resolution of 0.01° × 0.01° (Han & Wo 2018). The time range of data is from January to December 2020. The hourly precipitation is the checked data for QC and others are auxiliary.

For the convenience of data pre-processing, quality control reference domain is adopted to decide the valid spatial and temporal ranges of each data and make a best spatial-temporal match between the hourly rain gauge data and the other grid multi-source data. The hourly rain gauge data are accumulations over the hour and the rainfall is usually caused by the precipitation clouds around the checked station. According to the neighborhood principle, the valid spatial range of station data is a circular area with the given radius, and the valid spatial range of grid products is a rectangular area extending for given steps around the checked station located grid. For a given checked station, the quality control reference domain (QCRF) is defined as follows:

$$QCRF = (T_{RF}, S_{SRF}, S_{GRF}, R, N, n, T) \quad (1)$$

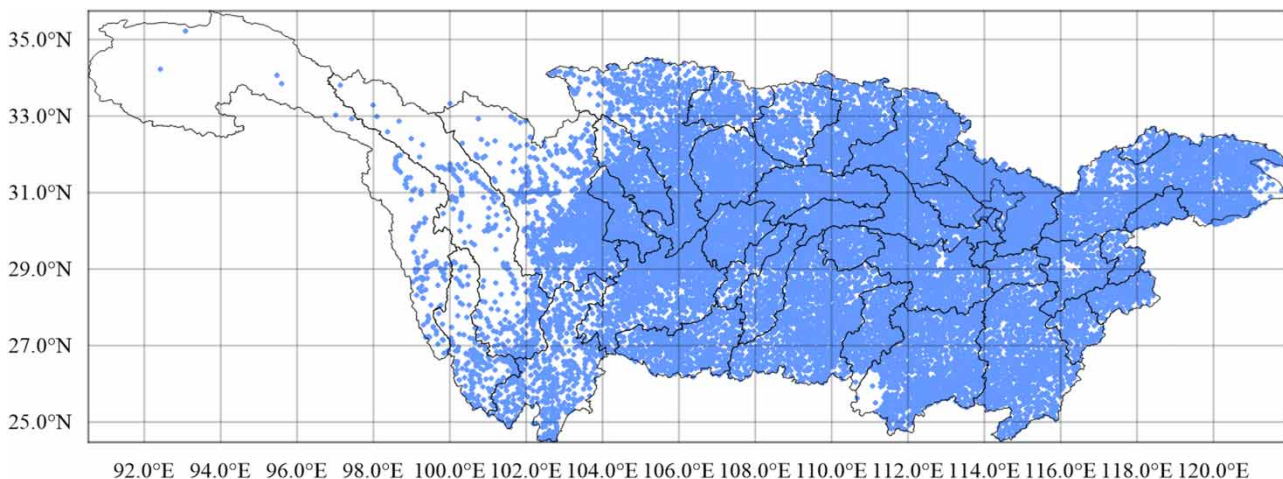


Figure 1 | Spatial distribution map of hydro-meteorological observation stations in the Yangtze River Basin.

where $T_{RF} = [HH_{-1} : 01, HH : 00]$ is the time range, HH is the hour of QC time, HH_{-1} is the previous hour; $S_{SRF} = \pi R^2$ is the spatial range of the station data, R is the given radius; $S_{GRF} = N \cdot S_{res}$ is the spatial range of grid products, S_{res} is the spatial resolution, $N = (2n + 1)^2$ is the total grid cells, n is the given extending steps; and T is the temporal frequency of grid product. It is clear that T_{RF} , S_{res} , and T are known for the given QC time and product, only R and n are uncertain. In this study, they are obtained by extensive experiments, and the results show that NMQC works best when n is 3 for FY-4A and CTA products, n is 1 for CR product, and R is from 50.0 to 70.0 with unit of km.

2.2. Definition of precipitation quality anomaly event

The concept of anomaly detection is introduced to improve the traceability of QC. The occurrence of anomalous precipitation data with large deviations from real precipitation is called a precipitation quality anomaly event (PQAE). Two categories of real-time and non-real-time events are summarily defined by comprehensively analyzing the causes, spatial-temporal distribution characteristics, and application sensitivity of a large number of anomaly data. Real-time events are gross errors in data, which have a great impact on the real-time applications such as weather forecasts and warnings, and usually can be labeled with a single time data. Non-real-time events are systemic errors in data, the elimination of which can provide high-quality data support for the non-real-time applications such as climate analysis and disaster assessment, and are needed to be labeled by analyzing the data changes over a period of time. NMQC focuses on the processing of real-time events, and non-real-time events would be discussed separately. Meanwhile, real-time events are subdivided into five types for more refined analysis, as presented in Table 1.

2.3. Calculation of QC-oriented parameters

2.3.1. Quality control factor

Quality control factors (QCFs) are designed based on station, radar, and satellite data with massive amounts of data exploration and analysis, which are sensitive to the quality confirmation of hourly rain gauge data. Station data are mainly used to design QCFs to characterize the precipitation extremes and spatial distribution around the checked station.

For radar data, although quantitative precipitation estimation (QPE) can be directly compared with the rain gauge data, it is usually calibrated by rain gauge data in real time to improve the accuracy of radar QPE (Koch *et al.* 2005; Xiao *et al.* 2008). Considering the coupling between rain gauge data and QPE product, only CR product is adopted, and QPE is directly estimated based on CR product according to the classical Z–R relationship (Fang *et al.* 2018).

For satellite data, CTT, CLM, CLT, and CTA products are used because the satellite rainfall estimation cannot be used directly in conjunction with gauge data, especially at hourly scale (Hughes 2006). Studies have shown that these parameters are closely related to ground precipitation (Liu *et al.* 2009; Han *et al.* 2011; Kim & Kwon 2011; Yuan & Hu 2015; Jin *et al.* 2018; Ombadi *et al.* 2021). The rainfall intensity has a good corresponding relationship with CTT, and CTT gradually decreases and its change range becomes more concentrated along with the increase of rainfall intensity, but CTT is less

Table 1 | Definition of real-time precipitation quality anomaly events

Event codes	Event names	Event definitions
CSPQAE	Clear sky precipitation quality anomaly event	Anomaly precipitation is observed on a clear day; there is no limit on the amount of precipitation.
PSPQAE	Pseudo small precipitation quality anomaly event	Anomaly precipitation is caused by equipment flipping without rain or fog, dew, frost, and snow gathering or melting; the amount of precipitation ranges from 0.1 to 0.3 mm.
IPQAE	Isolated precipitation quality anomaly event	Only the checked station has precipitation within a specific spatial range; the amount of precipitation is greater than 0.3 mm.
SLPQAE	Single larger precipitation quality anomaly event	Precipitation is significantly greater than that of all neighbor stations within a specific spatial range; the amount of precipitation is greater than or equal to 1.0 mm.
LPQAE	Larger precipitation quality anomaly event	Precipitation is significantly greater than that of neighbor stations or historical extremes within a specific spatial range; the amount of precipitation is greater than or equal to 1.0 mm.

sensitive to small precipitation. Thus, CLM, CLT, and CTA are introduced to enhance the robustness of satellite QCFs when precipitation is small or not obvious.

The set of QCFs can be denoted as $\{F_{ex}, F_{rto}, F_{cr}, F_{qpe}, F_{ctt}, F_{cta}, F_{dom}, F_{clr}\}$, which are calculated based on QCRF of the corresponding data or products. F_{ex} measures the history precipitation extremes around the checked station, F_{rto} represents the spatial frequency of stations where non-zero precipitation is observed, F_{cr} is the maximum of radar CR, F_{qpe} is the average of QPE for the Z-R relationship of different precipitation clouds, F_{ctt} is the minimum of CTT, F_{cta} is the average of CTA, F_{dom} is the cloud type with the highest proportion, F_{clr} represents the probability of clear sky, and the higher value corresponds to the lower probability of rainfall.

For the given checked station and QC time, t denotes the time order, i denotes the order number of grid cells, and QCFs can be described as follows

$$F_{ex} = f(lat, lon, alt, mth) \quad (2)$$

$$F_{rto} = \frac{N_{mb}}{N_{nb}} \times 100\% \quad (3)$$

where lat , lon , and alt are the latitude, longitude, and altitude of station, mth is the month of QC time, f is the extreme matching function and is used to obtain the monthly maximum hourly precipitation based on the defined QC parameter file, N_{nb} is the number of neighbor stations, and N_{mb} is the number of neighbor stations with rainfall. To reduce the boundary-effect influence on the stations located at the edge of the precipitation system, F_{rto} is calculated based on QCRF with $R < 50.0$ and QCRF with $R \in [50.0, 70.0]$ denoted by F_{rto_50} and F_{rto_70} , respectively.

$$F_{cr} = \max_{t=1}^T cr_t \quad (4)$$

$$F_{qpe} = avg(QPE_c, QPE_s, QPE_w) \quad (5)$$

where $cr_t = \max_{i=1}^N cr_t^i$ is the maximum CR at time t , cr_t^i is the CR of grid cell i at time t ; QPE_c , QPE_s , and QPE_w is the hourly QPE of the Z-R relationship of convective, stratiform, and warm cloud precipitation, respectively.

$$F_{ctt} = \min_{t=1}^T \min_{i=1}^N tbb_t^i \quad (6)$$

$$F_{cta} = \frac{\sum_{i=1}^N N_t^{cta}}{N} \times 100\% \quad (7)$$

where tbb_t^i is the CTT of grid point i at time t ; N_t^{cta} is the cloud total amount of grid point i .

The CLT product divides clouds into seven types: Clear, Water, Super Cooled, Mixed, Ice, Cirrus, and Overlap, which are abbreviated as *clr*, *war*, *scl*, *mix*, *ice*, *cir*, and *ovp*, respectively. Usually, precipitation will not occur in the area of Clear and Cirrus clouds. The valid values of CLM product include Cloud, Probably Cloud, Probably Clear, and Clear. And it is believed that precipitation will not occur in the area marked Clear.

$$F_{dom} = \arg \max_{s \in S} \sum_{t=1}^T N_t^s \quad (8)$$

$$F_{clr} = \min(F_{clm}, F_{clt}) \quad (9)$$

where $S = \{clr, war, scl, mix, ice, cir, ovp\}$ is the set of cloud types, s is the member of S , N_t^s is the amount of s grid cell at time t ; $F_{clm} = \min_{t=1}^T (\frac{N_t^{clr}}{N} \times 100\%)$, N_t^{clr} is the amount of clear grid cell at time t ; $F_{clt} = \min_{t=1}^T (\frac{N_t^{clr} + N_t^{cir}}{N} \times 100\%)$, N_t^{clr} is the amount of Clear cloud grid cell at time t , and N_t^{cir} is the amount of Cirrus cloud grid cell at time t .

2.3.2. Quality control decision index

To reduce the accuracy requirement for *QCFs* and improve the complexity of algorithms, the hierarchical fuzzy matching strategy is adopted to respectively further convert *QCFs* of radar and satellite into a quality control decision index, which is set into 10 grades from 0 to 9 to match with the pre-made hourly precipitation grade. It should be noted that the unit of hourly precipitation is millimeters (mm). In detail, 0 indicates that it may be a clear sky, 1 indicates it may be rain, 2~9 respectively indicate that the hourly precipitation is in the interval [0.1,1.0), [1.0,2.5), [2.5,10.0), [10.0,20.0), [20.0,30.0), [30.0,40.0), [40.0,60.0), and ≥ 60.0 . Besides, the default value M is set to improve the robustness of algorithms when data is missing, or the conversion is not feasible. The conversion rules of the quality control decision index are depicted as Figure 2, *QCDIS* and *QCDIR* denote the quality control decision index for satellite and radar, respectively. The diagnosis rules would be designed based on *QCDIS* and *QCDIR*.

2.3.3. Station weather type classification

Oriented to the requirements of QC, an empirical parameterized method is adopted to classify the weather type around the checked station by comprehensively analyzing multi-source *QCFs*. The station weather type (SWT) is divided into three types of *clear*, *rainless*, and *rain* as presented by Equation (10), where *clear* indicates that it is clear sky, *rainless* is no obvious precipitation, and *rain* is for precipitation, and the values of related *QCFs* are fitted with extensive experiments and application analysis. Different diagnosis rules are set for the same PQAEs according to station weather types.

$$SWT = \begin{cases} clear & \begin{cases} F_{rto_50} < 8.0\% \\ F_{cr} < 10.0dBZ \\ F_{ctt} \geq 278.0K \\ F_{ctr} \geq 60.0\% \\ F_{cta} < 40.0\% \\ F_{dom} \in \{clr, cir\} \end{cases} \\ rainless & \begin{cases} F_{rto_50} < 8.0\% \\ F_{rto_70} < 8.0\% \end{cases} \\ rain & other \end{cases} \quad (10)$$

2.4. The phased QC strategy for PQAEs

Because of uncertainties from radar and satellite remote-sensing measurements, the relationship is very complex between gauge rainfall and radar-satellite rainfall data; there is a high risk of misjudgment in use of radar and satellite data directly

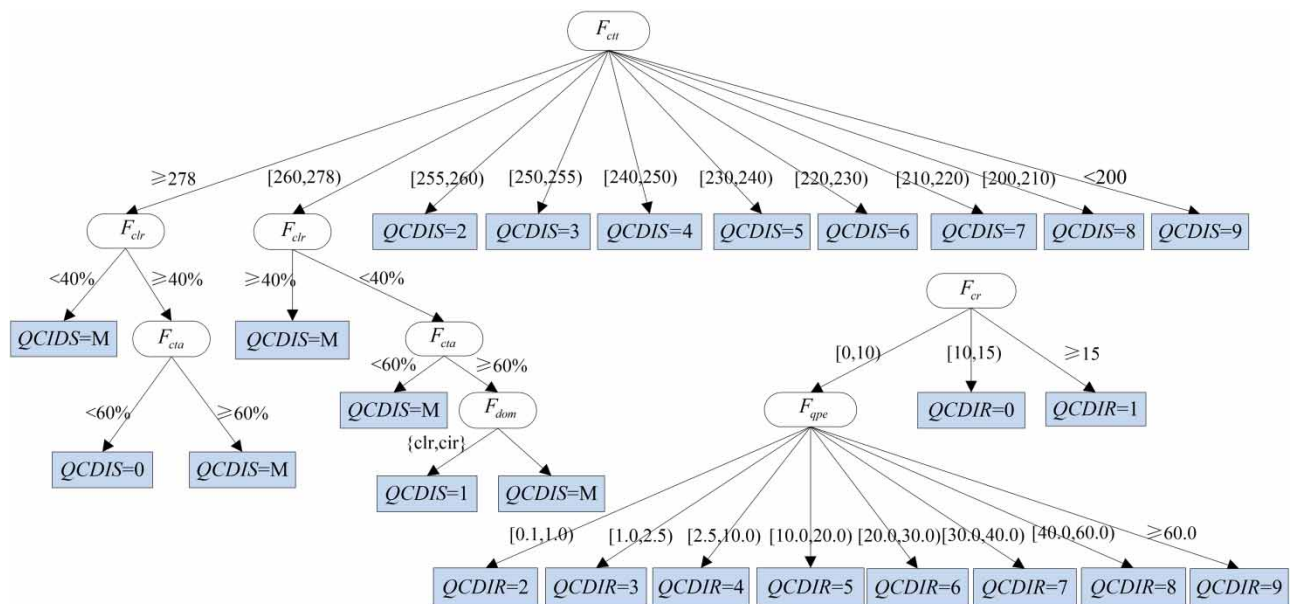


Figure 2 | Conversion rules of *QCDIS* and *QCDIR*.

to determine the quality of gauge data (Hughes 2006; Hill 2013; Qi & Zhang 2013). Thus, a phased strategy is designed to logically divide QC procedures into the two steps of PQAE detection and PQAE diagnosis to reduce the misjudgment risk. The phased QC strategy for PQAEs is depicted as Figure 3. First, the PQAE detection algorithm mainly uses rain gauge data to label as much of the suspected anomaly data as possible, which is similar to single-source QCs. Secondly, the PQAE diagnosis algorithm makes a comprehensive analysis of the suspected anomaly data based on QC parameters, and the suspected PQAEs are relabeled as normal or anomaly PQAEs. The data for normal PQAEs are available, and the data for anomaly PQAEs are suspicious or erroneous. The available, suspicious, and erroneous data are denoted by T, D and F, respectively. Suspicious and erroneous data are the final anomaly data labeled by NMQC, and suspicious data need to be further confirmed by data quality analysis experts. Hence, the labeled suspicious data should be as little as possible to make NMQC more automatic and reduce manual intervention.

2.4.1. PQAE detection algorithm

The PQAE detection algorithm is developed by combining spatial analysis with Isolation Forest to improve the efficiency of NMQC as shown in Figure 3(a). Spatial analysis is essential for rain gauge data QC. However, the efficiency will be limited when the amount of data is large because it works based on the calculation of neighbor stations' data. And it mainly focuses on the detection of anomaly data for heavy precipitation instead of small precipitation. Isolation Forest is an unsupervised machine learning algorithm that uses the mechanism of isolating outliers to perform anomaly detection (Ding & Fei 2013; Yao *et al.* 2022), which does not need to prepare the labeled training dataset in advance, and does not make assumptions about the probability distribution of the checked data. It is one of the most widely used algorithms in the field of anomaly data detection. First, Isolation Forest is adopted to quickly detect and label the suspected anomaly data. Secondly, the labeled suspected anomaly data are checked using the percentile-based spatial analysis method to obtain more refined spatial distribution parameters. Finally, comprehensive analyses restricted by business rules are made to classify the corresponding PQAEs.

2.4.2. PQAE diagnosis algorithm

The PQAE detection algorithm is implemented as shown in Figure 3(b). The data for Clear Sky Precipitation Quality Anomaly Event (CSPQAE) is directly confirmed as erroneous, because the corresponding SWT is *clear*, which has been comprehensively

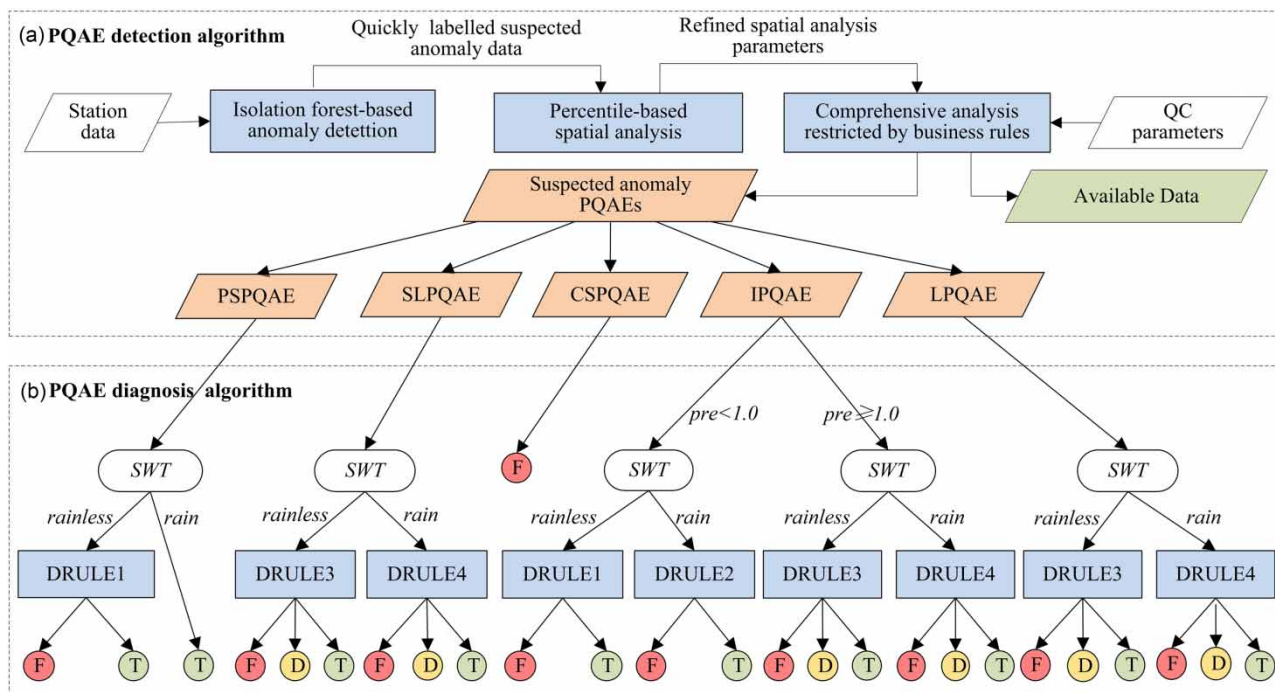


Figure 3 | Flowchart of the phased QC strategy for PQAEs.

assessed by multi-source data. For other PQAEs, considering that the recognition of small precipitation is more uncertain than heavy precipitation, the diagnosis rules are designed to vary with the hourly precipitation denoted with pre as follows: (1) when $pre < 1.0mm$ and there are no sufficient reasons for error, it is directly labeled with T as available. For instance, if SWT is *rain*, PSPQAE will be labeled with T; (2) when $pre \geq 1.0mm$ and there are no sufficient reasons for available or erroneous data, it is labeled with D as suspicious and confirmed by manual intervention if necessary. As defined above, 0 and 1 for $QCDIS$ are confirmed by multiple $QCFs$, which have higher accuracy compared with the corresponding values for $QCDIR$. On the contrary, when precipitation reaches a certain grade, the relationship between radar QPE and gauge precipitation is more accurate than that between CTT and gauge precipitation. Therefore, the priority of $QCDIS$ for small precipitation is higher than that of $QCDIR$, and the priority of $QCDIR$ for heavy precipitation is higher than that of $QCDIS$.

To facilitate the expression, the hourly precipitation grade is denoted by $PREGRD$. According to the above analysis, the threshold method is adopted to design four different diagnosis rules of DRULE1~DRULE4 on the basis of SWT , $PREGRD$, $QCDIR$, and $QCDIS$. Taking DRULE3 as an example, when SWT is *rainless* and $pre \geq 1.0mm$, it is used for comprehensive diagnosis of Isolated Precipitation Quality Anomaly Event (IPQAE), Single Larger Precipitation Quality Anomaly Event (SLPQAE), and Larger Precipitation Quality Anomaly Event (LPQAE). First, check if the value of $QCDIR$ is 0,1, or M, and if so, the data are labeled according to the pre-designed decision tables. Then, if not, the following steps are performed: (1) the parameters $a = PREGRD - QCDIR$ and $b = PREGRD - QCDIS$ are, respectively, defined and threshold c is given; (2) when $a < c$, if $PREGRD$ is 4 or 5, $QCDIR$ is 2, and $QCDIS$ is 0, the data are flagged with F as erroneous, otherwise they are flagged with T as available; (3) when $a \geq c$, if $QCDIS$ is 1 or M, the data are flagged with D as suspicious, or if $b < c$, they are flagged with T as available, otherwise they are flagged with D as suspicious.

3. RESULTS

NMQC is evaluated by analyzing the QC results of hourly rain gauge data from more than 24,000 hydro-meteorological stations in the Yangtze River Basin from January to December 2020. Data status flags are adopted to track and analyze the QC process of NMQC as presented in Table 2. During PQAE detection, data that are not labeled are available, which is flagged with C0, and the other labeled data are suspected anomaly data, which are flagged with C1. During PQAE diagnosis, C1 data are further confirmed as belonging to one of the three categories: available, suspicious, and erroneous, which are flagged with D0, D1, and D2, respectively. In addition, taking the hourly rain gauge data from June to August in Hubei Province as an example, a test dataset referred as $tSet$ is created to evaluate the performance of NMQC, where the anomaly dataset referred as $aSet$ is already labeled by data quality analysis experts.

On the basis of that, the detection ratio, hit ratio, false detection ratio, and miss detection ratio are used as metrics to evaluate the detection performance quantitatively, and the anomaly ratio for a single station is defined to understand the spatial distribution of anomaly data. The metrics are presented as follows:

$$R_c = \frac{N_c}{N_d} \times 100\% \quad (11)$$

$$R_s = \frac{N_s}{N_d} \times 100\% \quad (12)$$

$$R_e = \frac{N_e}{N_d} \times 100\% \quad (13)$$

Table 2 | Data status flags

Flag	Description
C0	Detected as true, data is available
C1	Detected as suspected anomaly data
D0	Detected as suspected anomaly data, diagnosed as available data
D1	Detected as suspected anomaly data, diagnosed as suspicious data
D2	Detected as suspected anomaly data, diagnosed as erroneous data

$$R_d = \frac{N_e + N_s}{N_d} \times 100\% \quad (14)$$

$$R_h = \frac{M_h}{M_d} \times 100\% \quad (15)$$

$$R_f = \frac{M_f}{M_a} \times 100\% \quad (16)$$

$$R_m = \frac{M_d - M_h}{M_d} \times 100\% \quad (17)$$

$$R_a = \frac{S_a}{S_d} \times 100\% \quad (18)$$

where R_c , R_s , R_e , and R_d are the detection ratios of C1, D1, D2, and D1 + D2 data, N_d is the total amount of QC data, N_c , N_s , and N_e are the total amounts of C1, D1, and D2 data; R_h , R_f , and R_m are the hit, false detection, and miss detection ratios of the anomaly data, M_d is the total amount of data in $aSet$, M_a is the total amount of anomaly data labeled by NMQC in $tSet$, M_h is the total amount of anomaly data labeled by NMQC as well as that appears in $aSet$, M_f is the total amount of anomaly data labeled by NMQC but that does not appear in $aSet$; R_a is the anomaly ratio for a given station, S_d is the total amount of QC data for the station, and S_a is the total amount of anomaly data labeled by NMQC for the station.

3.1. Detection performance of anomaly data

According to the statistical analysis of QC results, R_c , R_s , R_e , and R_d are 3.7, 0.03, 1.7, and 1.73%, respectively. It is clear that the detection ratio of anomaly data is reduced significantly after PQAE diagnosis. Concretely, the ratios of D0, D1, and D2 in C1 data are 53.74, 0.87, and 45.39%, respectively. Only 46.26% of the suspected anomaly data labeled during PQAE detection is ultimately confirmed as anomaly data during PQAE diagnosis, and the proportion of suspicious data is 1.73%. The monthly detection ratios of C1 data and D1 + D2 data are counted as shown in Figure 4. It can be seen that both the detection ratios for C1 and D1 + D2 data are lower in the summer half year (from April to September) than in the winter half year (from January to March, October to December), but the differences between them are higher in the summer half year than in the winter half year. In other words, NMQC has high false detection ratio in the summer half year during PQAE detection.

The metrics of hit ratio, false detection, and miss detection ratio for NMQC are verified based on the test dataset $tSet$. The results are listed in Table 3; M_d is 7,231, M_a is 7,215, M_f is 124, and M_h is 7,101. The statistical results show that the values of

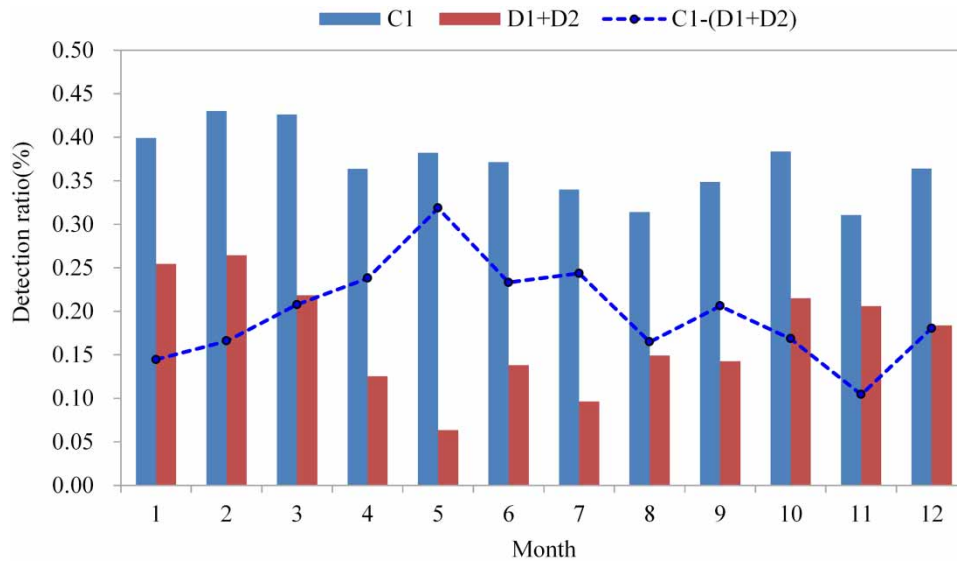


Figure 4 | Monthly detection ratio of C1 data and C1 + C2 data.

Table 3 | Verification results for NMQC on the test dataset *tSet*

Precipitation grades	M_d	M_a	M_f	M_h
≥ 1.0	413	403	13	390
[0.1,1.0)	6,818	6,812	101	6,711
Total	7,231	7,215	124	7,101

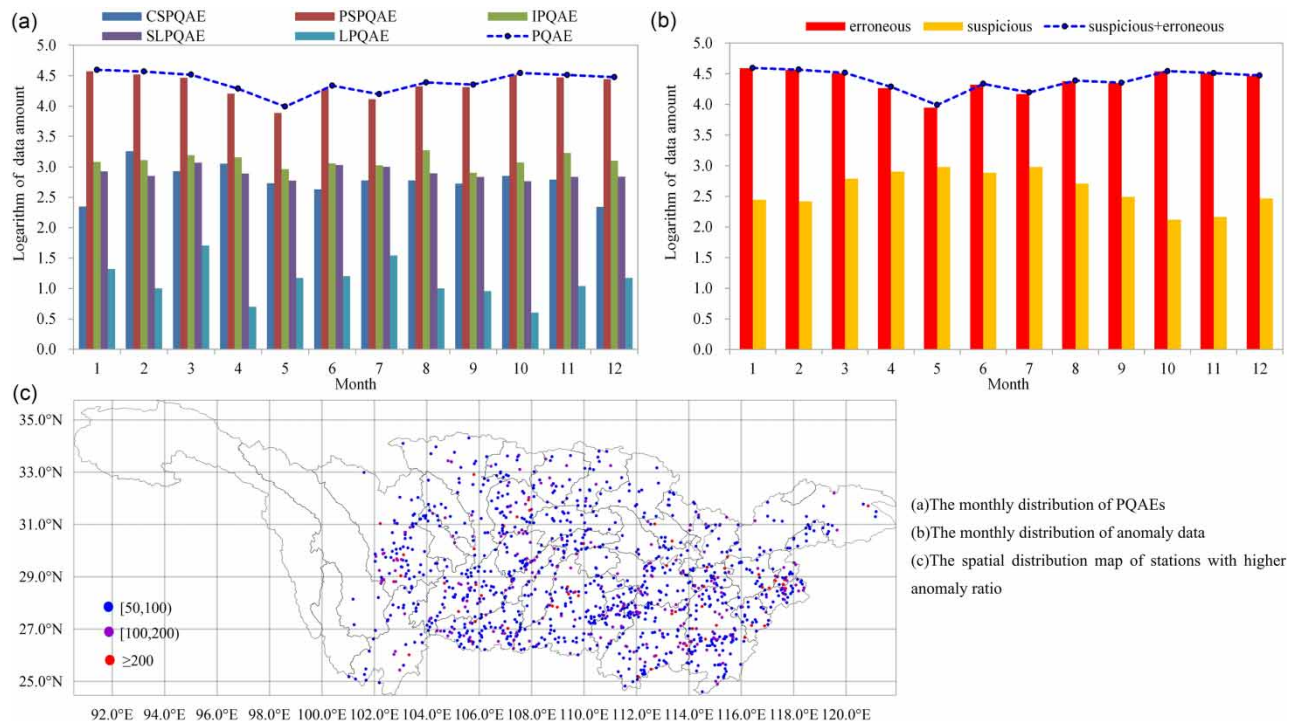
these metrics for anomaly data below 1.0 mm are better than that greater than 1.0 mm, and the total R_h , R_f , and R_m are 98.20, 1.71, and 1.80%, respectively.

3.2. Distribution characteristics of anomaly data

The distribution characteristics of anomaly data detected by NMQC are addressed to further verify the rationality of QC results and provide references for the application of the method. The main results are shown in Figure 5. First, the type differences and monthly change of anomaly data are explored by analyzing the distribution of PQAEs. Second, the proportion of erroneous and suspicious anomaly data and the temporal distribution of anomaly data are discussed to understand the automatic labeling ability of NMQC. Third, the spatial distribution of anomaly data is described to know the quality of hourly rain gauge data in different regions. Finally, the grades distribution of anomaly data is described in words to understand the quality of different grades of precipitation as well as the main grades of each PQA.

3.2.1. PQAEs distribution

To understand the event distribution of anomaly data, the ratio of different PQAEs is counted and the results show that the highest proportion belongs to PSPQAE with 89.52%, IPQAE is 4.84%, SLPQAE is 3.00%, CSPQAE is 2.58%, and LPQAE is the lowest with 0.06%. The monthly distribution of PQA is as shown in Figure 5(a). Due to the large difference in the proportion of different PQAEs, especially the proportion of PSPQAEs, which is much higher than others, the data amount is logarithmically converted to make the analysis clearer. It can be seen that the amount of PSPQAE is less in the summer

**Figure 5** | Distribution characteristics of anomaly data.

half year than in the winter half year; the amount of IPQAE is higher in March and August; the amount of SLPQAE is higher in March and from June to August; the amount of CSPQAE is higher from February to April; and the amount of LPQAE is higher in March and July.

3.2.2. Temporal distribution

Since the proportion of suspicious data is much lower than that of erroneous data, the data amount is logarithmically converted to better analyze the monthly change of anomaly data as shown in Figure 5(b). It can be seen that the anomaly data are more in the winter half year than in the summer half year. Meanwhile, the amount of erroneous data in the summer half year is less than that in other months, but the amount of suspicious data in the summer half year is more than that in other months. The main reason is that due to the influence of extreme weather such as local strong convection, it is more difficult to confirm the quality of data in the summer half year. The monthly change of anomaly data is consistent with that of erroneous data because the proportion of erroneous data is as high as 98.27%.

3.2.3. Spatial distribution

The statistical results show that the ratio of stations with anomaly data is 83.75%. Among these stations, the proportion of stations with $S_a < 10$ is 58.72%, $S_a \in [10, 50]$ is 35.57%, $S_a \in [50, 100]$ is 4.18%, $S_a \in [100, 200]$ is 1.21%, and $S_a \geq 200$ is 0.32%. The proportion of stations with $S_a < 50$ ($R_a < 0.57\%$) is 94.29%. Although most stations have anomaly data, the stations with high anomaly data ratio are relatively fewer. The spatial distribution of stations with higher anomaly ratio ($R_a \geq 0.57\%$) is as shown in Figure 5(c); it can be seen that the distribution of stations is relatively uniform and has no obvious regional characteristics except for areas with sparse station density. According to the analysis of stations with higher R_a , the main anomaly data are CSPQAE and PSPQAE, and these data are mainly the result of poor observation environment and instrument maintenance. Taking Yimencun station an example: it can be observed that it has the highest anomaly ratio R_a of 5.01%, and among the anomaly data, the ratio of PSPQAEs is 59.09%, CSPQAE is 38.41%, and others are 2.50%.

3.2.4. Grades distribution

The non-zero precipitation is analyzed to make a further understanding of the distribution of anomaly data in different precipitation grades. The results show that anomaly data are mainly concentrated at less than 1.0, which accounts for 94.36%. This is consistent with the fact that the observation samples of small precipitation are more. Moreover, the proportion of PQAEs to their own amount in different precipitation grades is counted to understand the precipitation grades distribution of different PQAEs. The results show that PSPQAEs are all in the $[0.4, 1.0)$, which is consistent with the definition; CSPQAEs occur at any grades, but are mainly concentrated below 1.0 accounting for 93.05%; IPQAEs are greater than or equal to 0.4, among which $[0.4, 1.0)$ account for 63.96%, and ≥ 1.0 account for 36.04%; SLPQAEs are equal to or greater than 1.0, among which $[1.0, 20.0)$ account for 90.68%, and ≥ 20.0 account for 9.32%; LPQAEs are greater than or equal to 1.0, among which $[1.0, 20.0)$ account for 6.18%, $[20.0, 60.0)$ account for 86.8%, and ≥ 60.0 account for 7.01%.

3.3. Case study

3.3.1. Pseudo small precipitation quality anomaly event

There are 102 stations showing PSPQAE at 10:00 on January 29, 2020 (Beijing time, the same as after) in the southeastern Sichuan and border areas of Chongqing, Hubei, and Guizhou ($28.30^\circ\text{--}32.30^\circ\text{N}$; $103.30^\circ\text{--}109.30^\circ\text{E}$), and these stations account for 1.7% of all stations in the region. The analysis of QC parameters shows that SWT is *rainless*, $QCDIS$ is 0, and $QCDIR$ is 0. The data are automatically confirmed as erroneous according to the DRULE1 diagnosis rule.

Some analyses are given to better understand PSPQAs as shown in Figure 6. The hourly average temperature of national stations from 27 to 31 January 2020 is depicted as Figure 6(a). According to the analysis, there was a large-scale precipitation process on 27 January 2020, which resulted in low temperature. The precipitation gradually stopped and the temperature slightly increased on the 28th. The weather cleared up on the 29th, the night radiation cooling effect and early precipitation provide favorable conditions for the emergence of fog, dew, and frost. The weather phenomenon data from 128 national stations on the 29th in the region are analyzed, and one or more phenomena within fog, dew, and frost occurred at 96 stations, and ice phenomena occurred at 18 high altitude stations. The spatial distribution of stations with PSPQAE and national stations recording fog, dew, frost, and ice phenomena is shown as Figure 6(b); it can be seen that the large-scale condensation phenomena occurred on 29 January 2020 in the region. Moreover, the hourly number of stations showing PSPQAEs from 21:00 on the 28th to 20:00 on the 29th is shown as Figure 6(c). There was no obvious change at night on

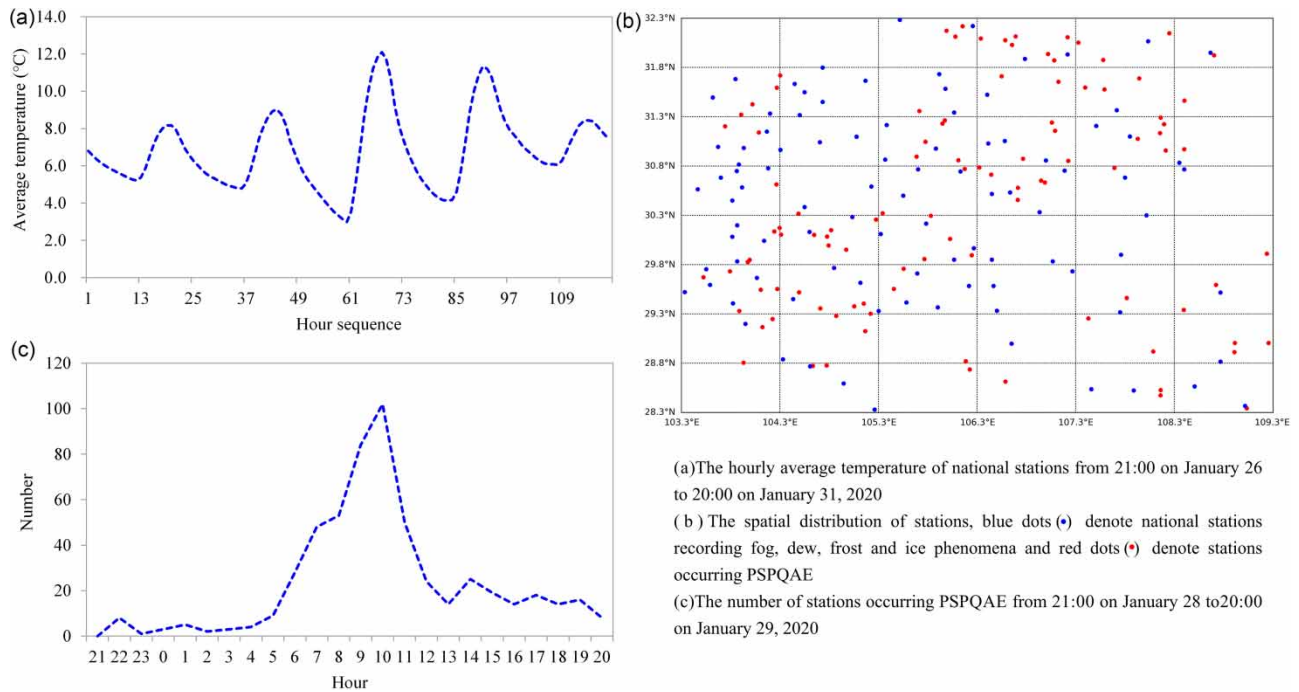


Figure 6 | The analysis of PSPQAEs in the southeastern Sichuan and border areas of Chongqing, Hubei, and Guizhou at 10:00 on 29 January 2020.

the 28th. As the temperature went up, the number rapidly increased on the morning of the 29th, reaching a maximum of 102 stations at 10:00, and then it quickly declined again. These facts are consistent with the formation and vanishing of fog, dew, and frost.

3.3.2. Clear sky precipitation quality anomaly event

Approximately 3.9, 3.2, and 2.3 mm of rainfall was observed at Loutai, Xiaojiagou, and Santai stations at 10:00 on 30 June 2020, respectively, which are detected as CSPQAEs. The analysis of QC parameters shows that the distance between each station is less than 10.0 km; F_{rto_50} is lower than 0.9%, and there is no precipitation at other neighbor stations; F_{rto_70} is 0.4%, and only one station has 0.5mm precipitation; SWT is *clear*. According to the multi-source data diagnosis rules, the data are automatically confirmed as erroneous. For further understanding, the spatial distribution of CR, CTT, and CTA around the station at 10:00 on 30 June 2020 are shown in Figure 7. According to the manual verification, the above

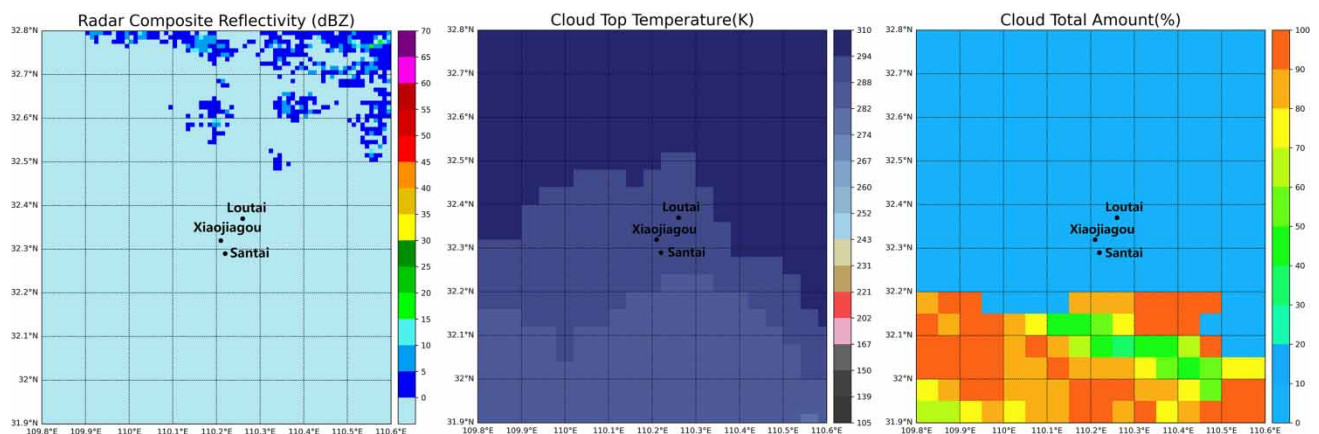


Figure 7 | The spatial distribution of multi-source data around Loutai, Xiaojiagou, and Santai stations at 10:00 on 30 June 2020.

precipitation is caused by station instrument maintenance. Due to the close distance and maintenance time between each station, the similar anomaly data are easily falsely detected by the single-source QCs. Fortunately, this can be avoided and the anomaly data are automatically labeled because of the usage of multi-source data.

3.3.3. Isolated precipitation quality anomaly event

About 23.0mm rainfall was observed at Guihua station at 14:00 on 28 August 2020, which is detected as IPQAE. It is labeled as C1 data during PQAE detection and automatically confirmed as available according to the DRULE3 diagnosis rule during PQAE diagnosis. The analysis of QC parameters shows that all the 52 neighbor stations within the 50.0km radius region have no precipitation, and all the 33 neighbor stations within the 50.0 ~ 70.0km radius region have no precipitation; F_{cr} is 43.5 dBz; F_{ctt} is 205.0 K; F_{dom} is Super Cooled Cloud; F_{clr} is 0.0%; F_{cta} is 100%; SWT is *rainless*, $PREGRD$ is 6, $QCDIR$ is 5, and $QCDIS$ is 8. The spatial distribution of CR, CTT, and CTL around the station at 14:00 on 28 August 2020 is shown in Figure 8. It can be seen that the precipitation of Guihua Station is stimulated by the small-scale strong convective cloud clusters. If there is no multi-source data diagnosis, this would be falsely labeled as anomaly data because of the uneven spatial distribution. The use of multi-source data can reduce the false detection ratio.

4. DISCUSSIONS

In this paper, NMQC is proposed for the hourly rain gauge data, which is a multi-source data quality control method. First, PQAEs are defined to improve the traceability management of the QC process. Second, several types of QC-oriented parameters are designed by the quantitative and fusion application of radar and satellite multi-source data to support the implementation of algorithms. Third, the phased QC strategy has been adopted to logically divide the QC procedure into the two steps of PQAE detection and PQAE diagnosis to reduce the misjudgment risk caused by the uncertainty from radar and satellite remote-sensing measurements. Finally, NMQC is evaluated by elaborating the QC results of hourly rain gauge data from more than 24,000 hydro-meteorological stations in the Yangtze River Basin from January to December 2020.

Overall, NMQC shows a good performance on the detection of anomaly data and has a strong ability to automatically label erroneous data. Compared with the operationally used single-source QC method (Ren *et al.* 2010), NMQC has a higher anomaly data detection ratio and a lower proportion of suspicious data detected (Section 3.1, R_s , R_e , and R_d for NMQC is 0.03, 1.7, and 1.73‰, respectively, and the proportion of suspicious data is 1.73‰). However, R_s , R_e , and R_d for the operationally used QC method is 0.44‰, 0.18‰, and 0.62‰, respectively, and the proportion of suspicious data is 70.97‰. The main reasons are as follows: (1) The operationally used QC method mainly focuses on the QC of heavy precipitation, but NMQC increases the QC of small precipitation, and the observation samples of small precipitation are much larger than that of heavy precipitation; (2) The confirmation of anomaly data is restricted by information resources for the single-source QC method because it is mainly based on spatio-temporal checks within the gauge network itself. But NMQC makes a comprehensive judgment from multiple perspectives by introducing satellite and radar data, which can greatly improve the automatic labeling ability for anomaly data.

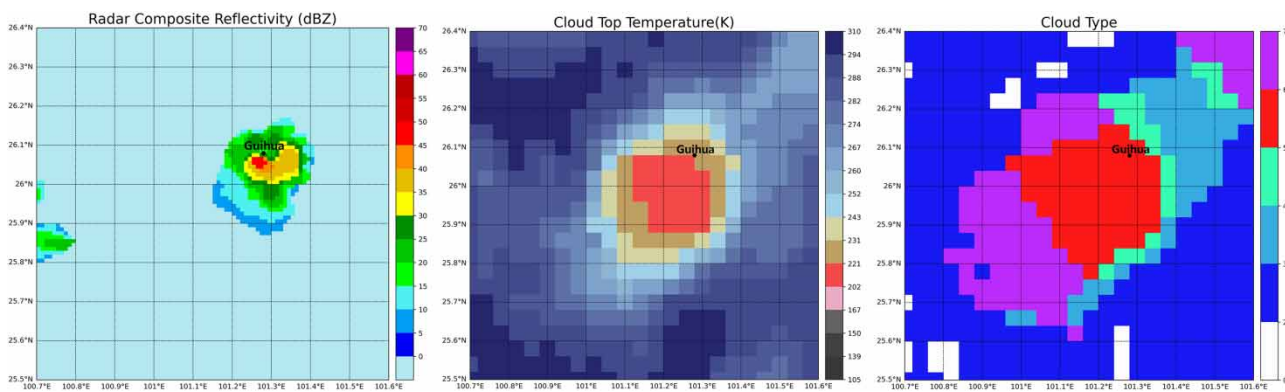


Figure 8 | The spatial distribution of multi-source data around Guihua stations at 14:00 on 28 August 2020.

Moreover, it should be noted that NMQC is not a fully automatic QC procedure compared with some existing multi-source QC methods; the suspicious anomaly data will be further confirmed by data quality analysis experts in the subsequent business operations. Qi *et al.* (2016) developed an automated gauge QC scheme based on the consistency of hourly gauge and radar QPE observations to benefit the making of gridded QPE products in radar coverage area. Notably, NMQC is designed for processing the gauge data itself. In other words, their application scenarios are not consistent. We believe that this is an important issue that must be carefully considered when designing QC procedures. Sha *et al.* (2021) present a supervised automated QC system using convolutional neural networks with grid precipitation and elevation analyses data as input for a sparse gauge station observation network; it is a meaningful exploration. Although manual QC is required to be performed on the raw gauge observations to create binary classification quality labels (good or bad) for the supervised training dataset, we think it is a possible future improvement direction for NMQC, which can be used as a better tool for making multi-classification labels.

Additionally, the results shown in Section 3 are consistent with the climatic characteristics of the Yangtze River Basin as well as the measurement and maintenance modes of rain gauges. Especially, NMQC has high false detection ratio in the summer half year during PQAE detection, and the anomaly data are more in the winter half year than in the summer half year after PQAE diagnosis. On the one hand, the rain and heat are observed over the same period in the Yangtze River Basin, and the local short-duration heavy precipitation and showers occur frequently in the summer half year (Cheng *et al.* 2022). The spatial distribution of such precipitation is usually similar to that of anomaly precipitation, and it can be easily falsely detected during PQAE detection. However, the authenticity of such precipitation can be determined during PQAE diagnosis by using satellite and radar data to eliminate false detection. On the other hand, the tipping bucket rain gauges are the main observation mediums, which have the inability to accurately measure solid precipitation (Colli *et al.* 2014; Martinaitis *et al.* 2015; Choi *et al.* 2022). There is snow or sleet weather in most areas of the Yangtze River Basin in the winter half year (Zhang *et al.* 2016). Although tipping bucket rain gauges are disabled in areas with prolonged snowfall of upper reaches, they are still used in middle-lower reaches during the sleet weather, which can cause much anomaly data. Meanwhile, data quality is also affected by the relatively loose rain gauge maintenance requirements because of the low probability of high-impact severe rainfall weather in winter in the Yangtze River Basin. For these reasons, it is suggested that the tipping bucket rain gauges precipitation data in the winter half year should be used cautiously in combination with the synoptic background. More in-depth research would be required for the QC of winter precipitation in the future.

Eventually, the present diagnosis rules are sensitive to the values of the relevant QC parameters, which are obtained through a large number of experiments. In the future, efforts will be made to develop more intelligent methods to set parameter values. Besides, although the hit, false, and miss detection ratios for NMQC are discussed on the labeled test dataset, the volume of the test dataset needs to be further expanded for more comprehensive evaluation.

5. CONCLUSION

In this study, we presented a novel quality control method for the hourly rain gauge data called NMQC to enhance the automatic labeling ability of anomaly data and the robustness of algorithms by the quantitative and fusion application of radar and satellite multi-source data in the Yangtze River Basin. The main conclusions are summarized as follows:

- (1) NMQC has a strong ability to label anomaly data automatically, which can greatly reduce manual intervention and shorten the impact time of anomaly data on applications. The detection ratio of anomaly data for NMQC is 1.73%, of which only 1.73% is suspicious data. And the hit, false detection, and miss detection ratios on the test dataset are 98.20, 1.17, and 1.80%, respectively.
- (2) The distribution characteristics of anomaly data detected by NMQC are consistent with climatic characteristics of the Yangtze River Basin as well as the measurement and maintenance modes of rain gauges. This implies the feasibility of NMQC. Specifically, the amount of different PQAEs varies greatly, with the highest proportion belonging to PSPQAE with 89.52%, and the anomaly data are mainly concentrated at less than 1.0 mm. The stations with high anomaly data ratios account for a small proportion and their spatial distribution is relatively uniform with no obvious regional characteristics. The anomaly data in the summer half year is less than that in the other months, but the amount of suspicious data in the summer half year is more than that in other months.

In summary, NMQC has a good performance on the detection of anomaly data with a lower proportion of suspicious data. Meanwhile, it also has a strong ability to automatically label erroneous data. Consequently, it can greatly reduce manual intervention and shorten the impact time of anomaly data. That is feasible for operational work.

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DATA AVAILABILITY STATEMENT

Data cannot be made publicly available; readers should contact the corresponding author for details.

CONFLICT OF INTEREST

The authors declare there is no conflict.

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